

Dirir Ahmed Ibrahim

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EDUCATION

University of Ottawa

Ottawa, ON

Bachelor of Science in Computer Science (Honors)

Expected Graduation, May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Programming Paradigms, Database Systems, Operating Systems, Computer Networks.

EXPERIENCE

Data Analyst Intern

Ottawa, Ontario

Senate of Canada

May 2026 – Aug 2026

- Analyzed, validated, and organized operational datasets using **Microsoft Excel**, ensuring data accuracy, consistency, and reliability for reporting and decision-making.
- Developed **Power BI dashboards and KPI reports** to transform transportation data into actionable insights and improve operational visibility.
- Utilized **SQL queries** to extract, filter, validate, and analyze operational data, supporting reporting, data quality, and business analysis.
- Prepared **briefing notes** and presented analytical findings and recommendations **at the director level**, supporting strategic and operational decision-making.

Quality Assurance Analyst Intern

Ottawa, Ontario

Senate of Canada

Sep 2026 – Present

- Developed and executed **Java-based JUnit unit and functional tests** to validate software functionality, identify and document defects, and ensure application reliability.
- Designed and executed **test cases and test scenarios** based on business and technical requirements to **verify software enhancements and system behavior**.
- Conducted **User Acceptance Testing (UAT)** by documenting test results, identifying defects, and validating software enhancements against business requirements.
- Analyzed **software requirements and system behavior**, documenting findings and recommendations to support **software quality and application reliability**.

PROJECTS

Net-Pulse | Go, TCP/IP Networking

- Built a high-concurrency network auditor to identify active services via TCP handshake validation.
- Leveraged **Goroutines** to execute parallel port probing, reducing discovery latency by over 70%.
- Developed a mapping engine to translate raw port data into human-readable system diagnostics.

File-Janitor | Go, System Automation

- Engineered a concurrent utility to automate directory organization and metadata-based sorting.
- Optimized I/O throughput via a **worker-pool pattern** for simultaneous high-volume data processing.
- Ensured 100% data integrity using atomic file operations and robust error-handling protocols.

TECHNICAL SKILLS

- **Foundations:** Data Structures, Algorithms, Object-Oriented Programming (OOP), Debugging, Unit Testing
- **Languages:** Java, Python, C++, JavaScript, TypeScript, C#, Go
- **Frameworks & Libraries:** Spring Boot, React, Node.js, Express.js, Django
- **Databases & APIs:** PostgreSQL, MongoDB, SQL, RESTful APIs, JSON
- **Tools & DevOps:** Git, GitHub, Linux/Unix, JUnit, Docker, AWS (EC2, S3), CI/CD